

NPI licensing in attitude reports with content-bearing nominals¹

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Abstract. Noun complement clauses as *in the claim that pigs are dirty* can host weak negative polarity items like *any* and *ever* that are licensed from the matrix clause. This property is shared with attitude verbs but contrasts with relative clauses to singular definite nouns, which is surprising in the light of recent predicate analyses of complement clauses. This puzzle has recently been observed by Sharvit (2024). In this paper, I show how a kind-based approach to attitudes can readily account for this.

Keywords: negative polarity items, attitude reports, kinds, attitudinal objects, downward entailment.

1. Introduction

In an increasing body of research, complement clauses are analyzed as predicates of content-bearing individuals rather than arguments of attitudes (e.g. Kratzer, 2006; Moulton, 2009; Elliott, 2017; Moltmann, 2020; Müller, 2020; Bondarenko, 2021). This move assimilates complement clauses to modifiers like adverbs on the clausal level. The treatment of attitudes becomes most transparent in the case of complex attitude reports. In those cases, a content nominal (CN) denotes an individual and the apparent complement clause is supposed to provide a predicate over said individual.

A straight-forward prediction that arises from this view is that complement clauses should behave like relative clauses.² This prediction, however, is not borne out. Sharvit (2024) observes that like ordinary complement clauses, but unlike relative clauses, CN clauses can host weak negative polarity items (NPIs) like *any* and *ever* under negation (or other elements that create a Strawson downward entailing environment) in the matrix clause. With Bondarenko and Elliott (2024a), I will call this Sharvit’s puzzle. It is illustrated in (1).

(1) Sharvit’s puzzle

- a. Layla didn’t claim/believe that Tonia ever went skydiving. *Verb complement clause*
- b. Layla didn’t believe the claim that Tonia ever went skydiving. *CN clause*
- c. # Layla didn’t believe the claim that Tonia has ever heard. *Relative clause*

This puzzle most obviously presents a problem for a relative clause or a predicate analysis of CN clauses as in (1b). This problem may hold regardless of how verb complement clauses in ordinary attitude reports are treated, which is a related but separate question.³ However, it

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²See Kayne (2008); Arsenijević (2009) for a syntactic analysis of so-called noun complement clauses as relative clauses.

³For instance, a modifier analysis is motivated by the optionality of complement clauses with verbs like *groan*, but many verbs like *believe* and *claim* have obligatory complements for which clauses alternate with regular nominal complements.

bears on the validity of compositional analyses for attitude verbs along the lines of *make the claim* or *have believe* (e.g. Moltmann, 2021, 2025; Müller, 2024). And the fact that CN clauses as in (1b) behave like verb complement clauses as in (1a) speaks in favor of a common analysis.

This paper is primarily concerned with the question of why sentences like (1b) can host weak NPIs in contrast to relative clauses. We will also analyze how they do so by exploring ways in which the entailment pattern can be accounted for. The analysis will be discussed for ‘equality semantics’ (Moulton, 2009, see e.g. Bondarenko and Elliott, 2024a for the term) and for object-based truth-maker semantics (cf. Fine, 2017; Moltmann, 2020), which is not straightforwardly monotonic. I will argue that both the downward entailing pattern and the contrast to ordinary relative clauses can be explained by assuming that CN clauses provide characterizing descriptions of attitudinal kinds. For more details on the general issue of the argument vs. the modifier approach to attitudes, the reader is referred to Sharvit (2024), and for subset vs. equality semantics to Bondarenko and Elliott (2024a).

In the following section, we briefly introduce the relevant background for NPI licensing, why it is a problem for relative clauses as in (1c), and why it should also be a problem for CN clauses as in (1b). In section 3, we discuss Bondarenko and Elliott’s (2024a) mereological approach, after which we discuss its problems and lay out a variant based on a more general principle. In section 5, I show that relative clauses and CNs actually pattern alike if we take into account kind-reference. I then proceed to show how the entailment pattern can be accounted for based on the kind-instantiation analysis of Müller (2024).

2. Background

2.1. NPIs in relative clauses with singular definites

Drawing on insights from the literature on negative polarity items (Lahiri, 1998; von Stechow, 1999; Guerzoni and Sharvit, 2007; Gajewski and Hsieh, 2014), Sharvit (2024) and Bondarenko and Elliott (2024a) assume that weak NPIs are licensed in environments that are Strawson downward entailing (SDE) but not Strawson upward entailing (SUE). A Strawson entailment $f(x) \Rightarrow_{ST} g(x)$ differs from regular entailment $f(x) \Rightarrow g(x)$ in that it only holds if $g(x)$ ’s presuppositions are satisfied. We will make the same assumption and I take the definition given by Sharvit (2024: 857, (41) and (42)) and repeat it slightly modified in (2).

- (2) a. An NPI α is acceptable only if α is dominated by an LF node β such that β is SDE, but not SUE, with respect to α ’s restrictor].
- b. δ is SDE with respect to γ iff for any g and γ' such that $\llbracket \gamma \rrbracket^g \Rightarrow \llbracket \gamma' \rrbracket^g$:
 $\llbracket \delta \rrbracket^g \Rightarrow_{ST} \llbracket \delta[\gamma/\gamma'] \rrbracket^g$
- c. δ is SUE with respect to γ iff for any g and γ' such that $\llbracket \gamma \rrbracket^g \Rightarrow \llbracket \gamma' \rrbracket^g$:
 $\llbracket \delta[\gamma/\gamma'] \rrbracket^g \Rightarrow_{ST} \llbracket \delta \rrbracket^g$

To exemplify this, let us look at a simple example. The sentence in (3) is SUE, but not SDE, as (3a-c) show, hence *any* is not licensed in (3d). The sentence with negation in (4) on the other hand is SDE, but not SUE, as (4a-c) show, hence *any* is licensed in (4d).

- (3) [Tonia ate a [fruit]].
- a. $\llbracket \text{apple} \rrbracket \Rightarrow_{ST} \llbracket \text{fruit} \rrbracket, \llbracket \text{fruit} \rrbracket \nRightarrow_{ST} \llbracket \text{apple} \rrbracket$
 - b. $\llbracket \text{Tonia ate an apple} \rrbracket \Rightarrow_{ST} \llbracket \text{Tonia ate a fruit} \rrbracket$
 - c. $\llbracket \text{Tonia ate a fruit} \rrbracket \nRightarrow_{ST} \llbracket \text{Tonia ate an apple} \rrbracket$
 - d. #Tonia ate any apple.⁴
- (4) [Tonia didn't eat a [fruit]].
- a. $\llbracket \text{apple} \rrbracket \Rightarrow_{ST} \llbracket \text{fruit} \rrbracket, \llbracket \text{fruit} \rrbracket \nRightarrow_{ST} \llbracket \text{apple} \rrbracket$
 - b. $\llbracket \text{Tonia didn't eat an apple} \rrbracket \nRightarrow_{ST} \llbracket \text{Tonia didn't eat a fruit} \rrbracket$
 - c. $\llbracket \text{Tonia didn't eat a fruit} \rrbracket \Rightarrow_{ST} \llbracket \text{Tonia didn't eat an apple} \rrbracket$
 - d. Tonia didn't eat any fruit.

Now, let us turn to relative clauses and definite descriptions. As (5) shows again, *any* is not acceptable within a relative clause that modifies a definite NP. The sentence is unacceptable despite the fact that *[any friends]* is in a SDE environment: If Layla didn't see the student who has friends then she also didn't see the student who has bad friends, for example, if we assume that $\llbracket \text{bad friends} \rrbracket \subseteq \llbracket \text{friends} \rrbracket$.

- (5) #Layla didn't see the student who has any friends.

The NPI in (5) is not licensed because singular definite descriptions are usually both SDE and SUE (Lahiri, 1998), due to their uniqueness presuppositions. Example (6) shows a standard lexical entry for the definite *the* with its uniqueness presupposition.⁵

- (6) $\llbracket \text{the} \rrbracket = \lambda P_{\langle e,t \rangle} : \exists! x[P(x)] . \iota x[P(x)]$

The problem is that if there is only one element in a set, then there can be maximally one element in its subset, and if there is an element in the subset, then it must be the same element. Example (7) exemplifies this: If the uniqueness presuppositions PS and PT of the two sentences S and T are satisfied, S and T mutually Strawson-entail each other.

- (7)
- a. S = Layla saw the student who has friends.
 - b. T = Layla saw the student who has bad friends.
 - c. PS = There is a unique student who has friends.
 - d. PT = There is a unique student who has bad friends.
 - e. Given PS and PT, the student who has friends = the student who has bad friends
 - f. $S \Leftrightarrow_{ST} T$

Bondarenko and Elliott (2024a) call this the *Uniqueness collapse*. I will rephrase it as in (8), which is tied to the definition of the definite in (6).

- (8) **Uniqueness collapse** (rephrased)

For any situation *s* and two sets *A*, *B* such that $A \subseteq B$, if there is a unique *x* in *s* such that $x \in A$ and there is a unique *y* in *s* such that $y \in B$, then $x = y$ in *s*.

Turning back to example (1c), if Tonia has only ever heard one claim in her life and that was yesterday, then (9a) and (9b) Strawson entail each other. Hence *ever* would be in an environment that is both SDE and SUE in (1c) and (9a).

⁴Excluding the free choice reading.

⁵We will not worry about maximality or maximal informativity for the moment.

- (9) a. Layla didn't believe the claim that Tonia has (#ever) heard.
 b. Layla didn't believe the claim that Tonia has heard yesterday.

We will now turn to NPI licensing with respect to attitudes and the semantics of content.

2.2. Analyzing attitudes

2.2.1. The modal analysis

In a standard, Hintikkan semantics, attitudes act as modals that universally quantify over a set of possible worlds, the set DOX^x of a person x 's doxastic alternatives (Hintikka, 1969). The semantics of a Hintikkan attitude can be represented as follows:

- (10) a. $\llbracket \text{Layla believes } \phi \rrbracket = \forall w \in DOX^{\text{Layla}}[\phi(w)]$
 b. For any individual $x \in D$: $DOX^x = \{w \mid w \text{ is compatible with what } x \text{ believes}\}$

This means that for x to believe a proposition ϕ , their set of doxastic alternatives needs to entail ϕ , i.e. $DOX^x \subseteq \phi$. Consequently, if $\phi \subseteq \psi$, then $DOX^x \subseteq \psi$, thus x believes ϕ entails x believes ψ , and x doesn't believe ψ entails x doesn't believe ϕ . Therefore we have the desired entailment pattern and since the attitude is not built with a singular definite description, there is no uniqueness presupposition involved and hence no uniqueness collapse.

However, the problem arises particularly for sentences with complex definite NPs like *the claim that she ever went skydiving* with which we are concerned in this paper. We will turn to this issue now in the context of the content individual analysis.

2.2.2. The content individual analysis

In a growing body of work, researchers have argued for an analysis of attitudes in which propositional content is predicated over a content-bearing individual (to name just a few, e.g. Moltmann, 1989, 2013, 2025; Kratzer, 2006, 2016; Moulton, 2009; Bogal-Allbritten, 2016; Elliott, 2017; Bondarenko, 2022). In this view, *that*-clauses act as modifiers of an individual, which could be an object or an event.⁶

For verbs, one of the motivations is the optionality of adding a *that*-clause to a manner-of-speech verb like *whisper* or *groan* as in (11). Here, the *that*-clause can be analyzed as an optional modifier which combines with the event argument of *groan* via predicate modification and provides the content of the groaning event.

- (11) Betül groaned (that life is unfair).

Content individuals become more obvious in the case of CNs like *rumor* in (12a) and the role that they play in complex attitude predicates like (12b-c) consisting of a light verb and a CN. In (12b-c), the 'attitude' is not in the verb or the event argument but is provided by the CN.

- (12) a. the rumor that pigs are dirty
 b. make the claim that pigs are dirty

⁶That is, an individual of type e or v . I will use the term content object only for type e individuals and the term content individual for both objects and eventualities.

- c. have the belief that pigs are dirty

There are (at least) two ways to define the content of a content individual, which Bondarenko and Elliott (2024a) call *subset semantics* and *equality semantics*. The former goes back to Kratzer (2006) and states that the content of a content individual entails a proposition. A version of it is represented in (13a). Equality semantics in the version presented here goes back to Moulton (2009). In (13) and (14), the subscripts in $CONT_{sub}$ and $CONT_{eq}$ disambiguate between the two versions. In the context of this paper, the important difference between these two versions is monotonicity. $CONT_{sub}$ is monotonic, (13b), whereas equality semantics is not monotonic, (14b), since the content is equated with the proposition.

(13) *Subset semantics*

- a. $\lambda p. \lambda x. CONT_{sub}(x) \subseteq p$
 b. For every $p, q \in P$ such that $p \subseteq q$: if $CONT_{sub}(x) \subseteq p$, then $CONT_{sub}(x) \subseteq q$.

(14) *Equality semantics*

- a. $\lambda p. \lambda x. CONT_{eq}(x) = p$
 b. For every $p, q \in P$ such that $p \neq q$: if $CONT_{eq}(x) = p$ and $CONT_{eq}(y) = q$, then $x \neq y$.

With this in place, we can model the semantics of (13a) and (14a) as follows, choosing equality semantics for the example.

- (15) a. $\exists e. \text{groan}(e) \ \& \ \text{Agent}(e) = \text{Betül} \ \& \ CONT_{eq}(e) = \llbracket \text{pigs are dirty} \rrbracket$
 b. $\exists! x [\text{rumor}(x) \ \& \ CONT_{eq}(x) = \llbracket \text{pigs are dirty} \rrbracket] . \iota x [\text{rumor}(x) \ \& \ CONT_{eq}(x) = \llbracket \text{pigs are dirty} \rrbracket]$

In her application of truth-maker semantics, a variant of situation semantics, to attitudes, Moltmann (2025) also adopts an equality semantics. Hence for present purposes, I will not make a formal distinction between a possible world and a truth-maker version of $CONT_{eq}$.⁷

Note that Kit Fine's truth-maker semantics in itself is not strictly monotonic because he defines the content of a sentence based on *exact* verification (Fine, 2017). Exact verification posits that a(n exact) verifier is wholly relevant to the truth of a proposition. Thus, bigger states that inexactly verify a proposition but include parts that lie outside the subject matter of a proposition, i.e. what the proposition is about, are not part of its set exact verifiers. Thus, the (exact) verifiers of the sentence *Joan buttered the toast with a knife* are not a subset of the (exact) verifiers of the sentence *Joan buttered the toast*.

It should be clear now why equality semantics (including truth-maker semantics) is not supposed to license NPIs: $CONT_{eq}$ provides neither upward nor downward entailment. The (non-)existence of an individual whose content is exactly p so far says nothing about the (non-)existence of an individual whose content is exactly q for $p \neq q$, whether p entails q or not. So far, the logical form for *making the claim that someone buttered the toast with a knife* does not entail *making the claim that someone buttered the toast*.

⁷ Given the bi-lateral structure of Fine's truth-maker semantics (Fine, 2017), Moltmann (2025: 121) defines a similar function *prop* as in (i). The second condition accounts for satisfiable objects that do not have violation conditions.

(i) For a sentence S , $\text{prop}(S) = \lambda d [\text{pos}(d) = \text{pos}(S) \ \& \ (\text{neg}(d) \neq \emptyset \rightarrow \text{neg}(d) = \text{neg}(S))]$.

For subset semantics, the problem is the uniqueness collapse that we are familiar with from the last section on relative clauses.

- (16)
- a. $S = \text{Betül made the claim that Tonia has friends.}$
 - b. $T = \text{Betül made the claim that Tonia has bad friends.}$
 - c. $\llbracket \text{Tonia has bad friends} \rrbracket \subseteq \llbracket \text{Tonia has friends} \rrbracket$
 - d. $PS = \text{There is a unique claim } x \text{ such that } \text{CONT}_{\text{sub}}(x) \subseteq \llbracket \text{Tonia has friends} \rrbracket.$
 - e. $PT = \text{There is a unique claim } y \text{ such that } \text{CONT}_{\text{sub}}(y) \subseteq \llbracket \text{Tonia has bad friends} \rrbracket.$
 - f. Given PS and PT, $x = y$, i.e. the claim that Tonia has friends is the claim that Tonia has bad friends.
 - g. $S \Leftrightarrow_{\text{ST}} T$

As a result, equality semantics never yields a SDE *nor* SUE environment and subset semantics always yields a SDE *and* a SUE environment, whether the matrix sentence contains a negation (or any other SDE element) or not. Either we have to find a way to get a SDE environment with equality semantics or we have to find a way to avoid the uniqueness collapse with subset semantics. There are ways around both issues. Let us turn Bondarenko and Elliott's (2024a) solution for equality semantics first.

3. Bondarenko & Elliot's mereological approach

Moltmann (2013, 2020, 2025) observes how we can talk about parts of attitudinal objects like claims, beliefs, and offers. Their part-structure becomes apparent in the use of *part of* or the adverb *partly*. The same predicates apply to parts, (17), parts of claims can be identified wrt. their content, (18b), and a part of whatever a *that*-clause in (18c) refers to can be true.

- (17)
- a. Part of Tonia's belief was true.
 - b. Tonia's belief was partly true.
 - c. Part of the offer still stands.
 - d. The offer was partly taken up.
- (18)
- a. He said that Tonia buttered the toast with a knife.
 - b. Part of his claim was that Tonia buttered the toast.
 - c. That Tonia buttered the toast with a knife is partly true.

Moltmann (2020, 2025) assumes parthood of attitudinal objects in terms of partial content in truth-maker semantics, which can be defined as follows with a state space S for a given model (see Fine, 2017: 569).

(19) **Partial content**

For two formulas A, B : B is a partial content of A iff every exact verifier of A contains an exact verifier of B and every exact verifier of B is contained in an exact verifier of A .

Bondarenko and Elliott (2024a) propose a solution for the non-monotonicity of equality semantics based on the mereological structure of content-bearing events and of propositions. They assume the following principle given in (20). I will assume $p' < \text{CONT}(x)$ to refer to partial

content as defined above.⁸

(20) **CONT satisfies Mapping to Subparts of the Input (MSI)**

$$p' < \text{CONT}(x) \rightarrow \exists x' < x [\text{CONT}(x') = p']$$

MSI in (20) guarantees that content objects like rumors and content eventualities like believings have parts with partial content. For phrases like *believe the rumor that S*, where we have two content individuals, a believing (in the sense of Bondarenko and Elliott, 2024a) and a rumor, Bondarenko and Elliott (2024a) also assume the following condition:

(21) **THEME satisfies Mapping to Subparts of the Output (MSO)**

$$e' < e \rightarrow \exists x' < \text{THEME}(e) [\text{THEME}(e') = x']$$

The idea is to assimilate sentences like (22a) to sentences like (22b). Krifka (1998) discusses sentences like (22b) wrt. their incrementality and proposes a mapping between events and their themes. MSI and MSO are versions of these mappings that are specific to content individuals.

- (22) a. Betül believed the rumor that Tonia buttered the toast with a knife.
b. Betül ate the apple.

Some attitude verbs, like *believe*, exhibit DP-to-CP entailment in the form of (23). Bondarenko and Elliott (2024a) list other verbs that exhibit DP-to-CP entailment: *deny, accept, doubt, memorize, trust, (dis)prove, validate, repeat*. These verbs contrast with others like *know, recognize, recall, forgot, discover, imagine, judge, regret, praise, and see*, which do not exhibit DP-to-CP entailment, cf. (24).

- (23) a. Betül believed the rumor that Tonia buttered the toast with a knife.
b. $\Rightarrow$ Betül believed that Tonia buttered the toast with a knife.

- (24) a. Betül knew the rumor that Tonia buttered the toast with a knife.
b. $\nRightarrow$ Betül knew that Tonia buttered the toast with a knife.

This indicates for cases like (23) that the content of the believing eventuality would be the same as the content of its theme. To ensure this, Bondarenko and Elliott (2024a) postulate the following condition:

(25) **Theme-Event Content Matching (TECM)**

- a. TECM holds of $P \in D\langle v, t \rangle$ iff:
 $P(e) \rightarrow (\text{CONT}(e) = \text{CONT}(\text{THEME}(e)))$

These conditions and postulates yield monotonicity for equality semantics: If there is a content individual x with a content p , then there is a partial individual x' with the partial content p' . And the content of a believing eventuality that has a theme will be the content of its theme.

⁸Bondarenko and Elliott (2024a) assume the notion in (i) for possible world semantics. However, since propositions are not reified but are assumed to be sets of possible worlds, I don't see how they can have parts in any other sense than subsets (which is not wanted here). I thus just assume a truth-maker version of MSI, similar to Bondarenko and Elliott (2024b) in a talk.

(i) Parthood for propositions
 $p \leq q$ iff p is semantically entailed by q (i.e., $p \supseteq q$).

In the following section, we will discuss some issues and refinements, before we go to a completely different, alternative approach in section 5.

4. Partial problems and attitude generation

There are important differences between the part structure of ordinary events like eatings and supposed attitudinal eventualities like believings as well as between ordinary objects like apples and content objects like claims.

Bondarenko and Elliott (2024a) mention that while parts of eating events are temporal parts, parts of believing eventualities are not. They suggest that parthood for believings "is crucially not grounded in the spatio-temporal properties of the eventualities in question" (Bondarenko and Elliott, 2024a: 28). As Moltmann (2013, 2020, 2025) has pointed out, parthood of attitudinal *objects* is based on partial content, never on temporal parts. Spatio-temporality, however, is one of the core characteristics of Davidsonian events. We may therefore question whether there are content-bearing believing eventualities at all. Due to incompatibility with the progressive and strong restrictions on their combination on adverbs, Davidsonians have often argued that (at least some) statives do not have an eventuality argument (see Davidson, 1967; Katz, 2003; Maienborn, 2007). Furthermore, CNs do typically not refer to eventualities and eventuality nouns are not associated with content (see Moulton, 2015; Moltmann, 2025):

- (26) a. I explained in under an hour that I was innocent.
 b. my explanation (*in under an hour) that I was innocent (*in under an hour)
 (adapted from Moulton, 2015: 316)
- (27) a. the belief/claim that pigs are dirty
 b. ??the speech act/mental state that pigs are dirty
- (28) a. This claim/belief is true.
 b. ??This speech act/mental state is true.
 c. ??This believing is true.
- (29) a. The belief/claim is that pigs are dirty.
 b. ??The speech act/mental state is that pigs are dirty.
 c. ??His believing is that pigs are dirty.

Thus, believings seem to have neither the spatio-temporal properties (e.g. temporal parthood) to be eventualities nor the ability to bear content or truth in order to be content individuals.

Let us look at the mapping from themes to partial events. In (30), the definite *the apple* refers to the whole apple, but due to the adverb *partly*, the sentence is perfectly felicitous in a context in which only part of the apple is eaten.

- (30) *Context: Betül ate part of an apple.*
 She partly ate the apple.

If the same principle is at play for content individuals, we expect (31) to be felicitous. However, it doesn't have the same level of acceptability that (30) has. The most available interpretation

of (31) without the context is one that refers to a temporal part: Layla was still speaking when she was interrupted. In this interpretation, *partly* yields a partial making-event and not a partial content individual.

- (31) *Context: Layla said that Tonia was in Italy.*
 ??Layla partly made the claim that Tonia was in Italy last week.

Another issue for the mapping between eventualities and content individuals concerns the difference between content objects and regular objects. When we say that someone ate an (or *the*) apple, we usually don't mean that they ate the whole apple, our statement would be compatible with leaving the core and the stem. Even if we say that they ate the whole apple, or in a clearer example, the whole pineapple, we don't necessarily mean that they ate all parts of it. But when we say someone believes a rumor, we assume that they believed all parts of it. Content objects have a greater mereological homogeneity than apples. Every part of a claim is *a* (whole) claim, but not every part of an apple is *an* apple.⁹

We arrived at a necessary condition: Every relevant part of a claim is also a (whole) claim. In fact, by making a specific claim (/assumption/holding a particular belief), we make several claims at once.¹⁰

- (32) a. Tonia went to NYC by train yesterday morning.
 b. Tonia went to NYC by train yesterday.
 c. Tonia went to NYC by train.
 d. Tonia went to NYC.
 e. Tonia left the house yesterday.

Each of these are full claims that we can refer to:

- (33) *Context: Layla said (32a).*
 a. Layla's claim that (32c)
 b. Layla made the claim that (32e).

It is therefore plausible to stipulate the following more general principle at play for attitudinal objects:

- (34) **Attitude generation**
 The instantiation of an attitudinal object generates instantiations of all content-entailed objects of the same type (= claim, belief, etc).

And define content-entailment as follows:

- (35) A content object *x* content-entails a content object *y*, if every exact satisfier of *x* is also an inexact satisfier of *y* and every exact satisfier of *y* is contained within a satisfier of *x*.

⁹They may be universally grinded *apple* though. Obviously, mass entities behave more homogeneously. In my intuition, *She drank the wine* would be acceptable if she didn't drink all of it, although *from the wine* would arguably be better.

¹⁰The intriguing question was brought up to me whether we are dealing with several different claims in (33) or just with different names for the same claim. Intuitively, one might say that Layla made *one* claim. However, one could imagine a pedantic argument for the view that Layla is making *several* claims, the claim that Tonia went to NYC, the claim that she did so by train and so on.

Downward entailment comes as an anti-generation inference in negative contexts: If Layla didn't make the claim that (32c) or (32e), they couldn't have made the claim that (32a) either, because (32c) and (32e) would have been generated by the instantiation of (32a). It is the same reasoning that applies for part-structure: if the part does not exist, the whole does not exist. The principle in (34) remains neutral wrt. whether it comes about by generation of parts. It could avoid some potential issues we discussed with the mereological mapping of content individuals and accounts for the fact that these parts can be referred to as full claims, not only as parts.

In the following section, however, we propose a different approach that is based on kind-reference.

5. A kind-based approach

5.1. NPIs with kind-referring definites

It has been observed that plural and kind-referring definites are not SUE and can license NPIs (see Hoeksema, 2008; Gajewski and Hsieh, 2014). In (36), the singular definite phrase *the student who has ever done this before* receives a generic interpretation. The NPI *even* is perfectly acceptable. Hoeksema (2008) argues that the lifting of the existential presupposition by the generic interpretation is responsible for the acceptability of NPIs in singular definites.

(36) The student who has ever done this before will not have problems with it.

I will assume that kind-referring singular definites refer to group-like taxonomic, or conceptual, entities (Dayal, 2004). In this view, kinds are not maximal sets of their members, but they inhabit our domain of entities just like regular individuals (see also Carlson, 1977).

Let us recall the problem with regular singular definites. An individual is unique wrt. to its membership in a given set in a given situation. The Strawson entailment adds another premise, the conclusion's presupposition that there is a unique member of a superset in a given situation. Due to the uniqueness collapse, the French student is then the same as the student in this situation and the student who has bad friends the same as the student who has friends, yielding Strawson upward entailment. Kinds, on the other hand, are unique by their nature. If they are unique globally, let's say in a maximal situation, a world, then they are unique in every smaller, partial situation. Thus, there is no uniqueness collapse of kinds in a particular situation and therefore also no Strawson upward entailment.

In general, however, taxonomic ordering can naturally license DE and UE inferences. For the kind predicate *extinct*, for instance, *The Lion is extinct* downward entails *The African Lion is extinct*. Hence, we have empirical, (36), and theoretical reasons to believe that a kind-based approach can potentially solve the issue of SUE and with it, the one of NPI licensing in sentences like (1b). We now turn to attitudinal kinds in particular.

5.2. Attitudinal kinds

Attitudinal objects in the sense of Moltmann (2020) come in kinds. Definite CNs with so-called complement clauses share with other kind-referring expressions their felicitousness in first-mention contexts and close apposition construction like *the claim that S, the wine Merlot*. Importantly, attitudinal CNs appear with kind predicates as in (37).

- (37) This belief/claim/rumor is common/widespread.

Attitude kinds are primarily defined in terms of content (Müller, 2024) and they are taxonomically ordered based on partial content. This is not entirely different from natural kinds. An Asiatic Lion distinguishes itself from a Lion by possessing additional qualities that also distinguish it from other subkinds of Lion.

- (38) An attitudinal kind A is taxonomically superordinate to an attitudinal kind B of the same type if A has a partial content of B.

Let us turn back to the main problem. The singular definite in (36) with a NPI involves a relative clause and is felicitous. So, what is the problem with the NPI in the relative clause in (1c), repeated below as (39)?

- (39) #Layla didn't believe the claim that Tonia has ever heard.

There are two issues with sentences like (39). First, relative clauses are not typical kind-definers.¹¹ As stated above in (38), attitudinal kinds are taxonomically ordered in terms of content. This makes it difficult to identify kinds of attitudinal objects (of a given type) in other ways. Second, the relative clause in (39) does not provide a good property to define a kind with its lowering operation triggered by *heard*.

Both issues can be resolved. We can use explicit kind-reference and we can use a characterizing sentence to define an *ad hoc* kind of claim. Then it should be clear that the CN refers to a kind defined by the relative clause. Our prediction is that the NPI is licensed if we make these changes to (1c)/(39). This is borne out:

- (40) a. Kate didn't make the kind of claim that could ever be harmful to anyone.
b. Kate doesn't accept the kind of claim that could ever be harmful to anyone.

Since we want the licensing to be due to the negation in the matrix clause (or any other licenser) and not only the kind reference, the sentences in (40) should become infelicitous if we remove the negation. This, again, is borne out:

- (41) a. #Kate made the kind of claim that could ever be harmful to anyone.
b. #Kate accepts the kind of claim that could ever be harmful to anyone.

We now turn to how exactly the entailment pattern is created.

¹¹Mendia (2020) argues that *ad hoc* kind reference is not readily available for singular definites with relative clauses. The generic interpretation of (36) above is not an issue, however.

5.3. Entailment pattern based on taxonomy

Kinds are taxonomically ordered and particular members are systematically related to their kind. As with sets, a member of a kind is also a member of the superordinate kind. A kind could be defined from its members. It is important to take into account, however, that we can refer to kinds with past but no current instantiations like *the Smilodon* and kinds that have never been instantiated in our world like *the Wolpertinger*. Taking up the notion of kinds as conceptual entities, I will go the other direction and assume an idealism-based version that defines membership from the kind. I will use superscripts as in x^k , y^o to visually distinguish between kind individuals and particular individuals.

- (42) a. $R(x^k, y^o)$ is a relation that holds between a kind individual x^k and a particular individual y^o iff y^o is a member of x^k (Carlson, 1977)
 b. $>_{TAX}$ is a taxonomic ordering such that $x^k >_{TAX} y^k$ iff x^k is taxonomically superordinate to y^k
 c. For all kind-individuals x^k, y^k such that $x^k >_{TAX} y^k$ and all particulars z^o :
 $R(z^o, y^k) \rightarrow R(z^o, x^k)$.

The crucial part (42c) states that every instantiation of the taxonomically lower kind is also an instantiation of the taxonomically higher kind.

In order to apply this to attitudes, I will adopt Müller's (2024) analysis of complex attitude predicates and assume $\llbracket \text{to claim that } S \rrbracket = \llbracket \text{to make the claim that } S \rrbracket$. A major difference of this approach from Bondarenko and Elliott's is that it does not involve content-bearing eventualities. Attitudes are decomposed into an attitudinal object and the contribution of a light verb. In the case of *make the claim*, *the claim* transparently refers to the attitudinal object and the light verb denotes a creation event. Crucially, *the claim that S* refers to a unique kind of claim defined by its content denoted by S .¹² According to the analysis, the phrase *make the claim* denotes the causation of an instantiation of an attitudinal kind. Since we are using the notions of attitudinal objects and partial content, it is most adequate to think of $CONT(x)$ as a function from objects to truthmakers in the sense of Fine (2017).¹³ The analysis is represented in a simplified version below in (43).

- (43) a. $\llbracket \text{the claim that } S \rrbracket = \iota x^k [\text{claim}(x^k) \ \& \ \text{CONT}(x^k) = S]$
 b. $\llbracket \text{make} \rrbracket = \lambda^k. \lambda e. \text{CAUSE}(e, \exists y^o [R(y^o, x^k)])$
 c. $\llbracket \text{make the claim that } S \rrbracket = \lambda e. \text{CAUSE}(e, \exists y^o [R(y^o, \iota x [\text{claim}(x^k) \ \& \ \text{fcont}(x^k) = S])])$

With (43c) and (42c), we get an UE inference. If someone made an instantiation of a certain claim-kind, then that individual is also an instantiation of the superordinate claim-kind. Note that according to this analysis, the sentence in (44b) would be predicted to be true in the context of (44a).

¹²For the sake of simplicity, I will neglect the uniqueness presupposition in the analyses below. As argued above, it is always satisfied when referring to kinds.

¹³Truthmaker semantics is bilateral. We are neglecting falsifiers here for the sake of simplicity, but adding them would not change the analysis or the outcome. We can also think of $CONT(x) = S$ as Moltmann's $\text{prop}(S)(x)$, which is cited in footnote 7.

- (44) a. Betül said that she went home by train at midnight.
 b. Betül made the claim that she went home at midnight.

Let us look at a negated example again.

- (45) a. Betül didn't make the claim [that she went home by train (= S)].
 b. Betül didn't make the claim [that she went home (= T)].

In (45), the claim-kind with content T is superordinate to the one with content S because T is a partial content of S. In (46), the premises in (46a-b) allow for the conclusion in (46c) with our condition on kind membership in (42c). With (43c) and (42c), and modus tollens, we get the desired DE inference. Any instantiation of the claim that S that Betül could have made would also be an instantiation of the claim that T. Therefore, if Betül didn't make an instantiation of the claim that T, she didn't make one of the claim that S.

- (46) a. $T >_{\text{TAX}} S$
 b. $\neg \exists e . \text{AGENT}(e) = \text{Betül} \ \& \ \text{CAUSE}(e, \exists y^o [R(y^o, \iota x [\text{claimk}(x^k) \ \& \ \text{fcont}(x^k) = T]])$
 c. $\neg \exists e . \text{AGENT}(e) = \text{Betül} \ \& \ \text{CAUSE}(e, \exists y^o [R(y^o, \iota x [\text{claimk}(x^k) \ \& \ \text{fcont}(x^k) = S]])$

We created exactly the right entailment pattern for complex attitude predicate with definite CNs. We did so within equality semantics creating the entailment not via part-structure but via kind-membership. Since kinds are universally unique, we also avoid running into the uniqueness collapse for standard subset semantics.¹⁴

The question now is how this approach carries over to cases like *believe the rumor*, in which we deal with two attitudes. First of all, we will make a similar assumption for *believe* as we made above for *claim*, namely that it can be decomposed along the lines of *have (the) belief* (see Moltmann, 2025). Thus, *believe* does not involve a content-bearing believing eventuality, but it involves a possessive-like relation to a belief object bearing the content. Second, we will assume that a personal belief is an instantiation of a belief-kind, in parallel to our analysis of *make the claim*.

Finally, we have to match the content of the two attitudinal objects, just like Bondarenko and Elliott (2024a) with the TECM postulate. However, many attitude verbs don't allow for this type of complement (**He claimed the rumor that S*) and not all of those that do exhibit DP to CP entailment. Out of the remaining ones, e.g. *believe*, *accept*, *validate*, *repeat*, the verb *believe* stands out in that it readily offers a respective content object, namely a belief, while it is less obvious that acceptance, validation, and repetition represent respective content objects. There are others that Bondarenko and Elliott (2024a) list, negative ones like *deny* and *doubt*, as well *prove*. For the negative attitudes, it is obvious that the content of what is denied, doubted, or rejected cannot be the content of the denial, doubt, or rejection. But also for *proof*, we find that the content of the proof (or a proving eventuality) is not the content of what is proven. Therefore, TECM does not hold in these cases. For this reason, namely that *believe* stands out

¹⁴While equality semantics seems intuitive for attitudinal kinds, it is conceivable to start out with a subset semantics that gets strengthened to equation by a maximality condition on the definite. This, however, is outside of the interest of this paper.

in this respect, I will build it directly into the meaning of *believe*.

Another thing that we want to ensure is that the attitude holder does not accidentally have a belief whose content matches the one of a certain rumor, but that they are also acquainted with that rumor, for example that they heard it. This is modeled as a presupposition in (47). However, we are not saying that the acquaintance event caused the belief. They could already have held the belief before or formed the belief at a later point in time. Since entailment comes about via instantiation of the belief, it does not matter in this case whether *the rumor* refers to a kind or a particular. The parts are given in (47) and applied to an example sentence in (48).

- (47) a. $\llbracket \text{the rumor that } S \rrbracket = \iota x [\text{claim}(x \ \& \ \text{CONT}(x) = S)] = \text{Rum}$
 b. $\llbracket \text{believe}_{\text{THEME}} \rrbracket = \lambda y \lambda x: x \text{ is acquainted with } y . \exists z [\text{POSS}(x, z) \ \& \ R(z, \iota b [\text{CONT}(b) = \text{CONT}(y)])]$
- (48) a. Kayla believed the rumor that S.
 b. *Kayla is acquainted with Rum* . $\exists x [\text{POSS}(\text{Kayla}, x) \ \& \ R(x, \iota b [\text{CONT}(b) = \text{CONT}(\text{Rum})])]$

Let us look at a negated example again:

- (49) a. Kayla didn't believe the rumor [that Betül buttered the toast with a knife (= S)].
 b. Kayla didn't believe the rumor [that Betül buttered the toast (= T)].

Since the content of the belief will match the content of its theme, the entailment pattern is the same as it is with a clausal complement. T is again ordered taxonomically higher than S, so if Kayla does not have a(n instance of the) belief that T, she also does not have a(n instance of the) belief that S.

- (50) a. $T >_{\text{TAX}} S$
 b. *Kayla is acquainted with the rumor that S* . $\neg \exists x [\text{POSS}(\text{Kayla}, x) \ \& \ R(x, \iota b [\text{CONT}(b) = \text{CONT}(\iota x [\text{claim}(x \ \& \ \text{CONT}(x) = T)])])]$
 c. *Kayla is acquainted with the rumor that T* . $\neg \exists x [\text{POSS}(\text{Kayla}, x) \ \& \ R(x, \iota b [\text{CONT}(b) = \text{CONT}(\iota x [\text{claim}(x \ \& \ \text{CONT}(x) = S)])])]$

In this analysis, we employ equality semantics to exactly define the content of a kind of attitude. However, we get standard subset semantics via kind-membership. In the composition, the kind is defined via equality semantics in the nominal part, but instantiation, and there for subset semantics, is introduced in the verbal part. This move allows us to get the desired downward entailment pattern that we would have for subsets without running into the SUE inference for singular definite CNs. While the kind is defined with an exact level of specificity, a member of a kind can be more specific. Just like a member of the kind *Lion* can be an Asiatic lion, or a member of the set *friends* can be a bad friend, a member of the kind *the claim that she went home* can be a claim with the content that she went home by train. We can now recapture and conclude.

6. Conclusion

In this paper, I have investigated a puzzle about the licensing of NPIs: Why do complex attitudes with nominal complements license weak NPIs the same way that attitude verbs with complement clauses do, but relative clauses do not license them? This question has been puzzling particularly in light of recent approaches to attitudes, according to which a noun complement clause provides a content predicate over a content individual in a similar fashion as relative clauses provide predicates over the individual denoted by the noun.

I have shown how Müller's (2024) kind-based analysis of attitudes readily accounts for the licensing of weak NPIs. For this, I built on the previous insight that generically interpreted singular definites are compatible with NPIs (Hoeksema, 2008). I argued that kinds, viewed as special individuals in our domain, are unique by nature and thus do not exhibit a uniqueness collapse that would lead to a Strawson upward entailment and would therefore ban NPIs. Entailment is created by the instantiation of attitudinal kinds and the fact that instantiations of a taxonomically lower kind are also instantiations of a higher kind. This is compatible with an equality semantics of content, where the content of the attitudinal kind is equated with the content of the embedded sentence. The account is based solely on attitudinal *objects* and does not involve content-bearing *eventualities*. I have adopted a truthmaker semantics-based view, but the crucial parts of the analysis are compatible with a semantics based on possible worlds.

In her original paper, Sharvit (2024) starts from the optionality of complement clauses with verbs like *groan* and *whisper* and their analysis within Neo-Davidsonian event semantics. This paper didn't bear on that issue, but the choice not to use content-bearing eventualities makes Kratzer's original analysis of these predicate impossible. Instead, constructions like *groan that S* would have to be analyzed as resultative constructions, where a groaning event results in a locutionary object with the content denoted by the complement clause. Moltmann (2025) observes that locutionary objects differ from attitudinal objects like beliefs in that they do not only have content-related but also form-related properties. This issue will be left to further research.

References

- Arsenijević, B. (2009). Clausal complementation as relativization. *Lingua* 119, 39–50.
- Bogal-Allbritten, E. (2016). *Building meaning in Navajo*. Ph. D. thesis, University of Massachusetts at Amherst.
- Bondarenko, T. (2021). The dual life of embedded CPs: Evidence from Russian *čto*-clauses. *Semantics and Linguistic Theory (SALT)* 31, 304–323.
- Bondarenko, T. (2022). *Anatomy of an Attitude*. Ph. D. thesis, Massachusetts Institute of Technology.
- Bondarenko, T. and P. D. Elliott (2024a). Monotonicity via mereology in the semantics of attitude reports.
- Bondarenko, T. and P. D. Elliott (2024b). On the monotonicity of attitudes: NPIs and clausal embedding. Talk held at the Complement Clauses Côte d'Azur colloquium, March 2024.
- Carlson, G. N. (1977). *Reference to Kinds in English*. Ph. D. thesis, University of Massachusetts, Amherst.

- Davidson, D. (1967). The logical form of action sentences. In N. Rescher (Ed.), *The Logic of Decision and Action*, pp. 105–148. Pittsburgh: University of Pittsburgh Press.
- Dayal, V. (2004). Number marking and (in)definiteness in kind terms. *Linguistics and Philosophy* 27, 393–450.
- Elliott, P. D. (2017). *Elements of clausal embedding*. Ph. D. thesis, University College London.
- Fine, K. (2017). *Truthmaker Semantics*, Chapter 22, pp. 556–577. John Wiley & Sons, Ltd.
- Gajewski, J. and I.-t. C. Hsieh (2014). Comments on negative polarity items in definite descriptions. In U. Sauerland and L. Crnič (Eds.), *The Art and Craft of Semantics: A Festschrift for Irene Heim*, pp. 181–198.
- Guerzoni, E. and Y. Sharvit (2007). A question of strength: On npis in interrogative clauses. *Linguistics and Philosophy* 30, 361–391.
- Hintikka, J. (1969). Semantics for propositional attitudes. In *Models for modalities: Selected essays*, pp. 87–111. Dordrecht: Reidel Publishing Company.
- Hoeksema, J. (2008). There is no number effect in the licensing of negative polarity items: A reply to guerzoni and sharvit. *Linguistics and Philosophy* 31, 397–407.
- Katz, G. (2003). Event arguments, adverb selection, and the stative adverb gap. In E. Lang, C. Maienborn, and C. Fabricius-Hansen (Eds.), *Modifying Adjuncts*. Berlin: Mouton de Gruyter.
- Kayne, R. S. (2008). Antisymmetry and the lexicon. *Linguistic Variation Yearbook* 8, 1—32.
- Kratzer, A. (2006). Decomposing attitude verbs. Talk given at the Hebrew University of Jerusalem.
- Kratzer, A. (2016). Evidential moods in attitude and speech reports. Talk given at University of Siena, University of Pennsylvania, and the University of Connecticut.
- Krifka, M. (1998). The origins of telicity. In S. Rothstein (Ed.), *Events and Grammar*, pp. 197–235. Dordrecht: Kluwer.
- Lahiri, U. (1998). Focus and negative polarity in hindi. *Natural Language Semantics* 6, 57–123.
- Maienborn, C. (2007). On Davidsonian and Kimian states. In I. Comorovski and K. von Heusinger (Eds.), *Existence: Semantics and Syntax*, pp. 107–130. Dordrecht: Springer Netherlands.
- Mendia, J. A. (2020). Reference to ad hoc kinds. *Linguistics and Philosophy* 43, 589–631.
- Moltmann, F. (1989). Nominal and clausal event predicates. *The 25th Chicago Linguistic Society (cls)* 25, 300–314.
- Moltmann, F. (2013). *Abstract Objects and the Semantics of Natural Language*. Oxford: Oxford University Press.
- Moltmann, F. (2020). Truthmaker semantics for natural language: Attitude verbs, modals and intensional transitive verbs. *Theoretical Linguistics* 46(3–4), 159–200.
- Moltmann, F. (2021). Truthmaker-based content: syntactic, semantic, and ontological contexts. *Theoretical Linguistics* 47(1–2), 155–187.
- Moltmann, F. (2025). *Objects and Attitudes*. Oxford: Oxford University Press.
- Moulton, K. (2009). *Natural selection and the syntax of clausal complementation*. Ph. D. thesis, University of Massachusetts, Amherst.
- Moulton, K. (2015). CPs: Copies and compositionality. *Linguistic Inquiry* 46(2), 305–342.
- Müller, K. (2020). Perception verbs and finite complement clauses. *Empirical issues in Syntax and Semantics* 13, 55–79.
- Müller, K. (2024). Making claims, decisions, and other things. *Proceedings of Semantics and*

Linguistic Theory 34, 666–684.

Sharvit, Y. (2024, Dec.). Standard and non-standard theories of attitudes and npis. *Proceedings of Sinn und Bedeutung* 28, 848–865.

von Stechow, P. (1999). NPI licensing, strawson entailment, and context dependencies. *Journal of Semantics* 16, 97–148.